Mittuniversitetet
MID SWEDEN UNIVERSITY

Försättsblad tentamen / Examination cover

2707183

Anonymitetskod / Anonymous code		
S - 0002 - AOW		
Kurskod / Course code: DT081A	Provkod/ Test code: T101	Tentamensdatum / Examination date: 2025-01-07
Kursnamn / Course name: Datateknik AV, Signal- och bildbehandling		
Provnamn / Test name: Tentamen		

Skriv din anonymitetskod på varje inlämnat papper
Write your anonymous code on each sheet submitted

Sätt ett kryss (x) för varje inlämnad uppgift
Use an x to indicate which questions has been submitted

Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes	Markera nedan med X / Mark below with an X		Poäng / Credit	Lärarens anteckningar / Teacher's notes
1	X	2		16			
2	X	1,5		17			
3	X	2		18			
4	X	4		19			
5	X	3		20			
6	X	4		21			
7	X	3		22			
8	X	5		23			
9	X	6		24			
10				25			
11				26			
12				27			
13				28			
14				29			
15				30			
Poängsumma / Points: 30,5			Betyg / Grade: C	Lärarsign./ Teachers sign			

Fylls i av tentamensvakt / To be filled in by the invigilator

Antal lösa blad/ No. of sheets submitted	Inlämnad tentamen / Submitted exam	Leg kontroll / Control identification	Sign. tentamensvakt / Sign. invigilator
11	–	OK	

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Försättsbladet skall alltid lämnas in även om ingen uppgift behandlats
Examination cover should always be submitted even if no questions are answered

Utskriven 2025-01-02 kl. 12:50:04

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Anonymitetskod/ Anonymus code: S-0002-A0W

Sidnr / Page no: 1

Lärarens anteckning / Teachers note:

Question :1 :-

a) Convolution is the multiplication of the input and the impulse response to an LTI system

→ Convolution can be calculated as

In frequency domain :

$$Y(f) = X(f) \cdot H(f) \;;$$

In time domain :

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\, h(t-\tau)\, d\tau \quad \text{(Continous)}$$

$$y[n] = \sum_{K=-\infty}^{\infty} x[K]\, h[n-K] \quad \text{(descrete)}$$

0,5

b) The function that transfers signal from time domain to frequency domain.

0

c) Response to an Impulse Input known as Impulse response. It considers short period of time.

0,5

d) Converting Continous signal to time descrete signal by spreading out over time is called sampling.

$$s[t] = s[nT]$$

0,5

Skriv ej på baksidan! Do not write on the other side!

e) Aliasing is conflicting of frequency while sampling. (Distortion of signal)
According to Nyquist theorem the sampling frequency must be $\geq 2 \cdot$ max frequency while sampling.

$$f_s \geq 2 f_{max}$$

Lärarens anteckning / Teachers note:

0,5

/2

Lärarens anteckning / Teachers note:

Question : 2 :-

a) Its an IIR system because it feedback in it. This means also it depends on previous output as well as present and past Input.

1

b)

$$y(n)=x(n)+4x(n-1)+2x(n-3)+0.4y(n-1)-0.2y(n-3)$$

2

3

Lärarens anteckning / Teachers note:

Question 03 :-

a) As we are giving Impulse input to system the response will be Impulse response. 1

b) $y(0) = x(0) + y(0-1) + y(0-2)$

$y(0) = x(0) = ?$

$y(1) = x(1) + y(1-1) + y(1-2)$

$y(1) = x(1) + x(0) = ?$

$y(2) = x(2) + y(2-1) + y(2-2)$

$y(2) = x(2) + 2x(0) + x(1) = ?$ 1

$y(3) = x(3) + y(3-1) + y(3-2) = x(3) + y(2) + y(1)$

$= x(3) + x(2) + 2x(0) + x(1) + x(1) + x(0) =$

$y(3) = x(3) + x(2) + 3x(0) + 2x(1) = ?$

$y(4) = x(4) + y(4-1) + y(4-2)$

$y(4) = x(4) + y(3) + y(2) = ?$

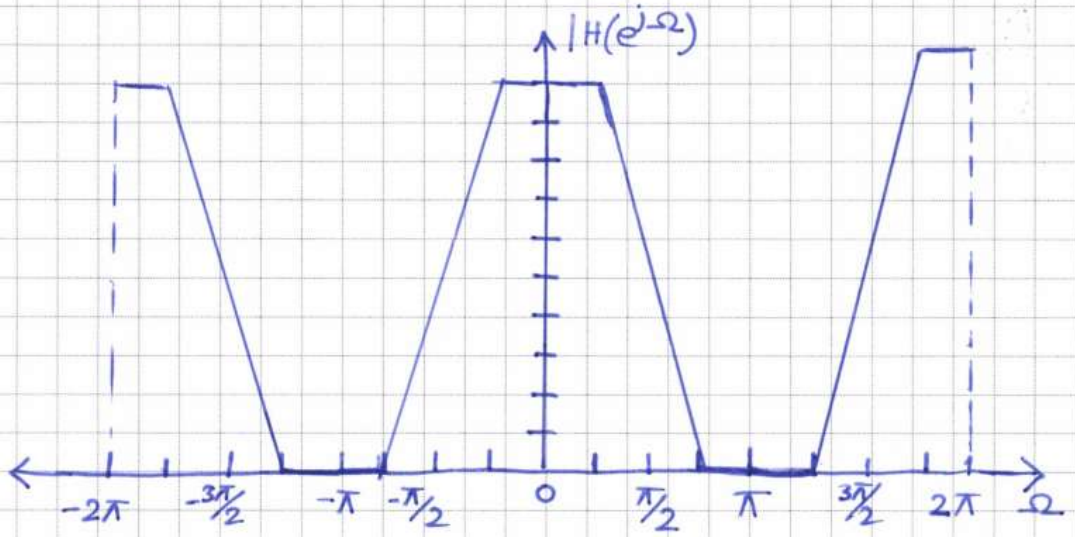
Evaluation missing

2

Lärarens anteckning / Teachers note:

Question : 4 :-

a)



2

b) As we can see in the figure we have $\frac{3}{4}\pi$ and $f_s = 100KH_2$

And for normalizeed frequency range it is

$$\frac{\Omega}{2\pi} f_s = \frac{3}{4}\pi \times \frac{100}{2\pi}$$

2

lowest sampling $f = \frac{3}{8} \times 100 = 37.5\ KH_2$

According to Nyquist shannon thcorem

$f_s \geq 2 f_{max}$.

4

Lärarens anteckning / Teachers note:

2

Question 5 :-

a) → Amplitude :

$$|Y(f)| = |X(f)| \cdot |H(f)|$$

→ Phase :

$$\angle Y(f) = \angle X(f) + \angle H(f)$$

also written as $\arg Y(f) = \arg X(f) + \arg H(f)$

b) After Considering the figures or applying product of amplitude we can see in

→ $|X(f)|$ is 4 for $0 \le f \le 10$
→ $|X(f)|$ is decreasing for $10 < f \le 30$
→ $|X(f)|$ is 0 for $f \ge 30$

And

$|H(f)|$ is 10 for $0 \le f \le 10$
$|H(f)|$ is leaneraly decreasing for $10 < f < 30$

By Considering figures and

$$|Y(f)| = |X(f)| \cdot |H(f)|$$

$$|Y(f)| = \begin{cases} 40 & \text{for } 0 \le f \le 10 \\ 80 - 4(f-10) & \text{for } 10 < f \le 30 \\ 0 & \text{for } f > 30 \end{cases}$$

Phase shift will be 0 for all value.

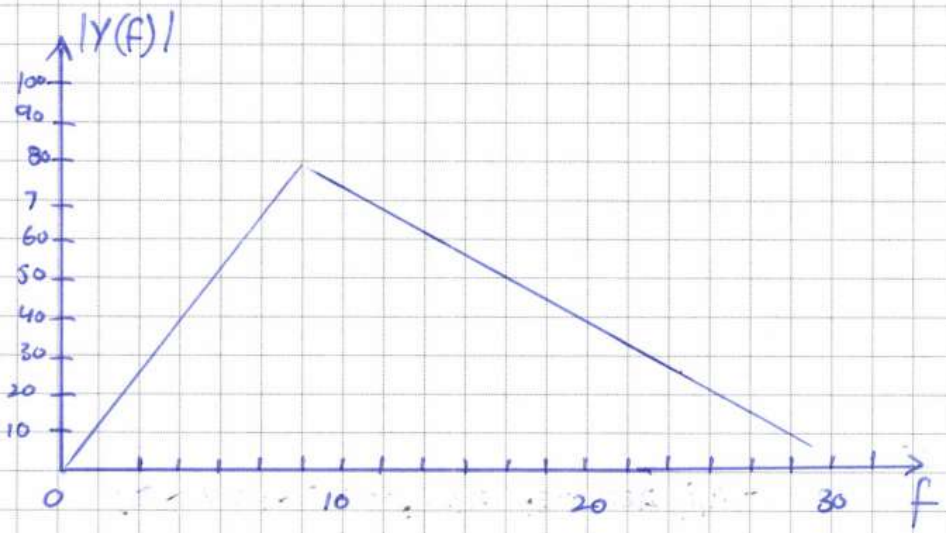
$$\arg(H(f)) = 0 \quad \forall f$$

Lärarens anteckning / Teachers note:

1

3

Amplitude Spectrum:



Lärarens anteckning / Teachers note:

Question NO: 06 :-

a) Figure (iii) is acheiving best frequency seperation because it has narrow main lob and frequency components are showing correctly.

2

b) Figure (iv) is best in minimizing the spectral leakage.
As we can see in the figure main lob is more extended and there is no shorter frequencies left on the sides.

2

4

Lärarens anteckning / Teachers note:

Question 7 :-

As we can see clearly the mid point is high and corners (edges) of the diagram are at low. This means higher frequencies are at lower level and lower frequencies are at higher level. So, if we are allowing the system to pass only lower frequencies it means low pass filter is applied. as edges represent higher frequencies.

3

Lärarens anteckning / Teachers note:

Question 8:-

→ 1=iv

Brightness kernal is applied as all points are low except Middle point. we can say in identity kernal we have adeleed more brightness to imge by making it 2. 1

→ 2=v)

Vertical Kernel is applied thats why it shows verticle edges. 1

→ 3=(ii)

Horizontal Kernal is applied thats why we can see horizontal edges. 1

→ 4=(i)

Box blur Kernel is applied as all indexis are very low thats why Image is blurred. 1

→ 5=(iii)

Sharpress kernal is applied we can see in the Image is comptly sharp due to this kernal. 1

5

Lärarens anteckning / Teachers note:

Question No: 09

By Assuming some the points like middle one's and corners we can find the 3×3 Kernal.

if we Consider $x(1,1)$ and $y(1,1)$ also $x(2,3)$ and $y(2,3)$ also other middle points we will have 0.5 for Kernal $h(1,1)$

By Considering $x(0,1)$ and $y(0,1)$ and appling point spread function the value for $h(0,1)$ will be 0.125

By Considering $x(0,0)$ and $y(0,0)$ we can see $h(0,0)$ can be 0

Also same for $x(0,2)$ and $y(0,2)$ the value will be 0 for $h(0,2)$

if we conside $x(4,3)$, $x(2,1)$ and $y(4,3)$ and $y(2,1)$ we can see the Kernal value for $h(2,1)$ will be 0.125

And then we can calcute value for $h(1,2)$ by considering points in $x(1,2)$ and $y(1,2)$ that will also be 0.125

$h(2,0)$ will be zeroo as it is outside and in others does not effect when we consider so as middle point.

By Considering also other points we will get Kerned

$h =$

i \ j	0	1	2
0	0	0.125	0
1	0.125	0.5	0.125
2	0	0.125	0

3×3 matrix.

6